

Extending the Senses of Ubiquitous Devices

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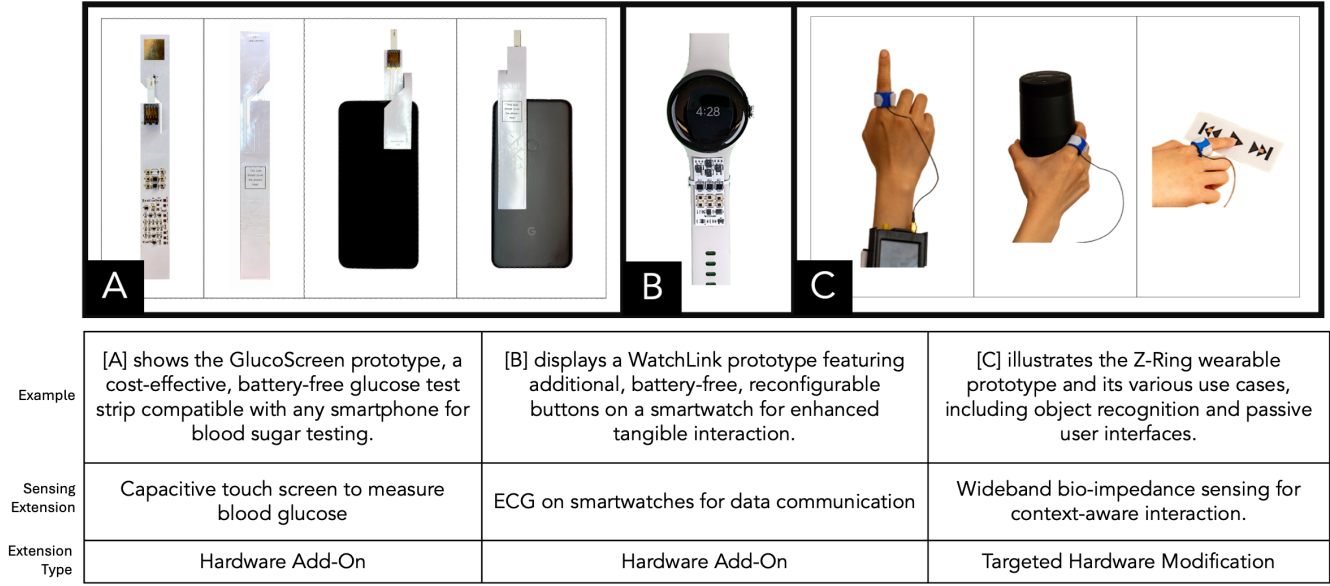


Figure 1: Enhanced side-channel sensing techniques used to broaden the functionality of ubiquitous devices, detailing the types of sensing expanded and the specific enhancements applied.

ABSTRACT

Modern smart devices are equipped with numerous sensors that can serve new functions beyond their original design when repurposed through side-channel sensing. Side-channel sensing involves leveraging existing sensors in unconventional ways to detect subtle signals and gather information beyond their original intended purpose. This work explores the untapped potential of side-channel sensing to enhance the functionality of existing smart devices, particularly in addressing niche user needs. I present two approaches: The first technique involves low-cost, low-power sensor add-ons that users can attach to their devices, enhancing functionality without any internal modifications to the device. I showcase GlucoScreen, a smartphone add-on that leverages the capacitive touchscreen for blood glucose monitoring, and WatchLink, which allows users to connect external sensors to their smartwatches via the ECG interface. The second technique focuses on manufacturers making targeted upgrades to existing hardware to improve functionalities

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at reduced time and cost. I present Z-Ring, a ring wearable that expands the bandwidth of bio-impedance sensing to enable micro-gesture interactions and object identification. Our findings highlight the potential of side-channel sensing to create more personalized and adaptable technology.

CCS CONCEPTS

• **Human-centered computing** → **Ubiquitous and mobile devices; Interaction devices**; • **Hardware** → **Sensor devices and platforms**.

KEYWORDS

Sensing, Health Monitoring, Wearable Input

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1 INTRODUCTION

In today's era of pervasive technology, we are inundated with smartphones, smartwatches, and other smart devices intended to enhance our health, navigation, communication, security and user

interaction. These devices host an array of sensors, like IMUs, microphones, cameras, and fingerprint scanners, that augment both device functionality and the overall user experience. In short, our devices have become indispensable tools in our lives.

Though smart device sensors offer a wide range of functionalities, they are often designed with the needs of the general population in mind. Manufacturers prioritize features that appeal to a broad audience due to market demand and cost considerations. As a result, individuals with niche requirements or specific use cases can be overlooked, unable to fully leverage the potential of their devices to meet their unique needs.

This thesis aims to bridge this utility gap by empowering *users to integrate custom sensors into their existing smart devices* to broaden their capabilities and *manufacturers to adapt their current hardware to enhance device functionality* without necessitating a complete redesign. I achieve this objective by harnessing the capabilities of existing sensors, unlocking their full potential through what I refer to as **side-channel sensing**. This technique involves repurposing existing sensors in unconventional ways to detect subtle signals from user interactions and gather information that addresses needs beyond the sensors' original intended purpose.

By leveraging this often-overlooked data, we can forge novel applications, such as enabling smartphones to perform blood glucose testing and adding battery-free health sensors that monitor temperature, UV light intensity, and breath alcohol. We can also enrich user experiences by enabling context-aware subtle input for augmented reality (AR) and virtual reality (VR) scenarios or by adding battery-free buttons to enhance smartwatch interactions. Side-channel sensing thus maximizes the potential of existing sensors, broadening the scope of device capability to new applications that were previously unattainable with the original designs.

My research delves into two primary approaches to achieve an optimal balance between increased functionality and practicality in side-channel sensing (see 1). The first approach centers on *developing cost-effective and energy-efficient hardware add-ons that can be easily integrated with existing devices*. The second investigates *hardware modifications that can enhance the capabilities of existing sensors without requiring a complete device redesign*. Figure 2 compares these two approaches, highlighting their respective trade-offs.

2 LOW-COST AND LOW-POWER HARDWARE ADD-ONS

This approach revolves around designing and testing hardware add-ons that prioritize three key factors: (1) cross-device compatibility, (2) affordability, and (3) low power consumption. First, these add-ons should be versatile, working seamlessly with various commercially available devices. This ensures broader user adoption and avoids limitations specific to certain device models. Second, cost is a crucial factor. Users should be able to readily afford these add-ons, making the technology accessible to a larger audience. Finally, minimizing power consumption is essential. The add-ons should have minimal impact on battery life, avoiding the need for frequent charging and ensuring a smooth user experience. By focusing on these priorities, we can create side-channel sensing add-ons that are practical, user-friendly, and widely adoptable.

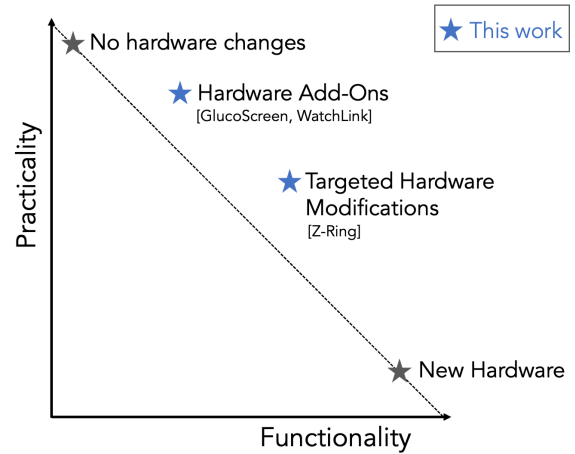


Figure 2: The trade-off between functionality and practicality in side-channel sensing. Applications requiring minimal hardware changes offer the most practicality, while those requiring new hardware may offer greater functionality but at the cost of reduced practicality. The ideal scenario lies in the middle ground, where functionality can be enhanced without sacrificing ease of use. Hardware add-ons and modifications offer a practical way to achieve this balance.

Hardware add-ons offer a compelling solution with several key benefits. By offloading complex data processing to the smart device itself, the add-on can remain simple in design and cost-effective to produce. This division of labor ensures efficient utilization of resources – the add-on collects the raw data, while the smart device handles the heavy lifting of analysis. Further, the add-ons can act as 'bridges,' making data from external sources accessible to the smart device. This opens access to a wider range of functionalities and unlocks the potential for side-channel sensing to interact with a much richer ecosystem of data streams, going beyond the limitations of the built-in sensors.

I introduce below two such add-ons that I developed, GlucoScreen and WatchLink.

2.1 GlucoScreen

An example of a low-cost, low-power add-on is GlucoScreen[12], a smartphone-based solution for prediabetes screening (prototype shown in Figure 3). This device attaches to a smartphone, eliminating the need for a separate glucose test strip reader. GlucoScreen enables blood glucose testing directly on the phone, leveraging the widespread availability of smartphones to reduce screening costs significantly.

Early detection of prediabetes is crucial since it promotes timely lifestyle modifications that can potentially reverse the condition and prevent the onset of full-blown diabetes[7]. However, current screening methods, like lab tests and glucometers, are often expensive and inaccessible [3, 8], leading to a significant number of undiagnosed and untreated cases. GlucoScreen addresses this issue

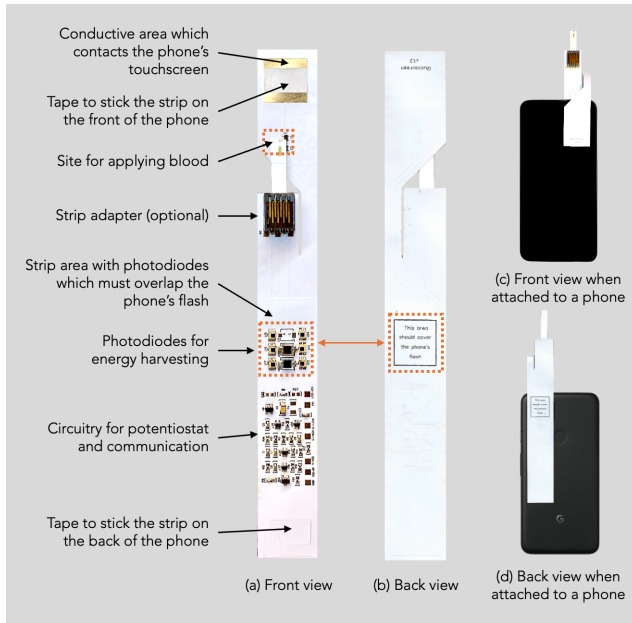


Figure 3: The GlucoScreen prototype and how it attaches to a smartphone.

by providing an accessible and low-cost screening option, empowering individuals to take control of their health and make informed decisions about seeking professional medical care. At-home screening also lowers the barrier to action for getting screened, thereby increasing the likelihood of more people getting screened.

GlucoScreen leverages the smartphone’s capacitive touchscreen as a communication channel. It generates touch events electrically from the test strip to the phone, encoding data in the timing of these events. This novel approach requires minimal components, resulting in a cost-effective and power-efficient design. Its minimal power requirements can be easily fulfilled by harvesting energy from the phone’s flash, eliminating the need for batteries and extending the device’s shelf life. The smartphone processes the touchscreen events, making the results available for immediate readout and storage.

Rigorous testing in both laboratory and clinical settings has demonstrated that GlucoScreen’s performance is on par with commercially available glucose monitors, further solidifying its potential as a valuable tool for prediabetes screening.

2.2 WatchLink

Another illustration of this approach is WatchLink[11], an add-on sensor platform for smartwatches. WatchLink, as shown in Figure 4, facilitates the connection of external sensors to smartwatches in a low cost, power-efficient manner.

WatchLink focuses on delivering specialized features for smartwatch users with specific needs, features that manufacturers often omit to maintain mass-market appeal. For instance, field workers might need UV sensors to monitor and mitigate skin damage from

sun exposure, while people in areas with high air pollution could benefit from air quality monitoring devices.

Integrating such niche sensors into widely produced wearables presents challenges, including high costs and complex integration processes that could limit their mainstream appeal. This gap in the market opens up possibilities for cost-efficient, specialized add-on sensors that leverage the existing capabilities of smartwatches, such as their display and processing power, while also maintaining low power consumption and cost. The primary challenge remains developing a practical and affordable way to connect these sensors to the watches.

In response, WatchLink proposes a novel method that uses the existing electrocardiogram (ECG) hardware on smartwatches to receive data from external sensors, enabling low-power communication with minimal and inexpensive components. Our approach involves transmitting sensor data through frequency modulation of voltage variations that the ECG hardware, designed to detect minute voltage potentials from heart activity, can then capture. Compared to traditional communication technologies such as Bluetooth Low Energy (BLE) or WiFi, this method offers significant advantages for add-on sensors, namely, it consumes significantly less power during active communication [1, 2, 5, 9] and requires only a few low-cost modulation components, making it about ten times cheaper than BLE implementations.

To showcase its feasibility and versatility, four distinct sensor modules were developed: a UV light sensor, a body temperature sensor, external buttons for smartwatch control, and a breath alcohol sensor. These modules successfully communicate with popular smartwatch models like the Apple Watch Series 9 and Google Pixel Watch 2 through the ECG interface, highlighting the practicality and potential of this communication technique.

3 HARDWARE MODIFICATIONS FOR ENHANCED CAPABILITIES

Hardware add-ons offer a compelling solution for expanding functionalities, but some applications might demand modifications to the existing hardware on the smart devices themselves. While this approach may seem less practical compared to simple add-ons, it holds significant advantages over a complete system redesign or developing entirely new components from scratch.

The key to the success of this approach lies in reducing costs and development time. Modifying existing hardware is generally less expensive and time-consuming than a complete overhaul. This translates to lower Bill of Materials (BOM) costs for manufacturers, making the technology more commercially viable. Since existing components form the base, development cycles can be shorter compared to designing entirely new hardware from scratch.

Furthermore, this approach helps preserve user familiarity. Extensive hardware modifications can drastically alter the form and function of a device, potentially alienating users. By modifying existing hardware, we can introduce new functionalities without significantly changing the user experience. This ensures a smoother transition for existing users who are already familiar with the device’s form and operation. In essence, we can enhance functionalities without sacrificing user comfort and ease of use.

I built Z-Ring as a proof-of-concept of this approach.

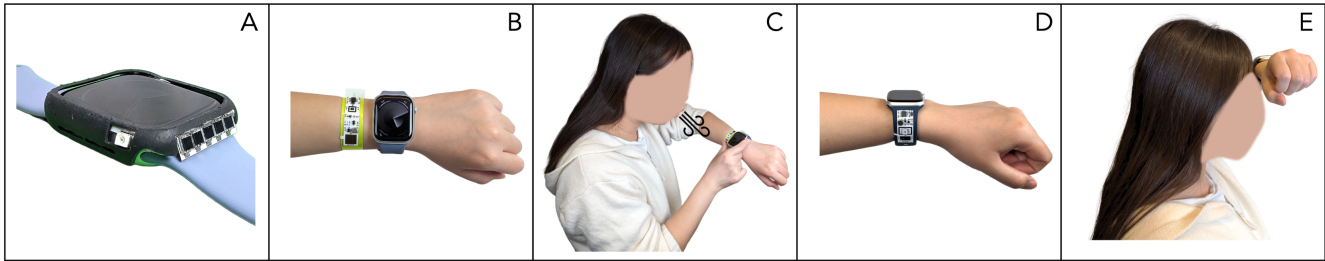


Figure 4: WatchLink lets users add sensors to commodity smartwatches via the ECG interface. Sensors can be (A) integrated into a watch case or (B) worn as a companion accessory, like this alcohol sensor. (C) A user is blowing on the worn breath alcohol sensor, and ECG communication is coupled through the user’s fingers. (D) Sensors can also be added to watch straps, like this body temperature sensor, which (E) can be touched to the forehead for measurement.

3.1 Z-Ring

Z-Ring[10], a ring designed for subtle and contextual user input, illustrates how targeted hardware modification can enhance functionality. Unlike traditional smart health devices that use bio-impedance sensing within a narrow bandwidth for stress assessment, Z-Ring explores the megahertz (MHz) range. This broader bandwidth allows the human hand to act as an antenna [4, 6] and its electromagnetic properties to vary with hand pose, motion, and contact with objects.

Z-Ring can detect subtle finger movements by measuring bio-impedance on this broader bandwidth, enabling micro-gesture interactions. Furthermore, since electrical signals can travel through the hand to external objects, the ring can identify objects being held or surfaces being touched without any modification to those objects. Z-Ring also incorporates a set of interfaces, including buttons, a 1D slider, and a 2D trackpad, designed to create distinct impedance profiles for different interactions. This design allows the ring to determine the point of contact without requiring additional electronics or batteries. Moreover, Z-Ring can distinguish wearers based on the unique frequency signatures generated by anatomical differences, adding a personalized aspect to its functionality.

A comprehensive user study with 21 participants confirmed Z-Ring’s effectiveness in gesture recognition, touch detection on external surfaces, object identification, and user recognition, demonstrating the transformative impact of targeted hardware enhancements on smart device utility.

4 FUTURE WORK AND CONCLUSION

This thesis demonstrates the potential of side-channel sensing to expand the capabilities of smart devices, particularly in addressing niche user needs. By repurposing existing sensors to detect subtle signals and gather information beyond their intended purpose, we unlock new possibilities for personalized and adaptable technology.

Glucoscreen, a cost-effective and power-efficient hardware add-on, showcases the potential of side-channel sensing by leveraging a smartphone’s capacitive touchscreen for blood glucose monitoring. This exemplifies how existing hardware can be creatively utilized to address specific user requirements. WatchLink further exemplifies this approach by leveraging existing ECG sensors in smartwatches to enable the connection of additional low-power sensors, opening up new possibilities for health monitoring and personalized data

collection. Z-Ring illustrates the power of side-channel sensing through its use of wide-bandwidth bio-impedance sensing. This innovative approach enables micro-gesture interactions and object identification, showcasing the potential for enhancing user interaction and personalization.

Side-channel sensing offers immense opportunities. My future research will focus on expanding its applications to other sensor modalities, developing advanced signal processing techniques, leveraging recent advancements in machine learning for increased performance, and integrating them with emerging technologies. Additionally, addressing privacy concerns and ensuring robust security mechanisms will be crucial as this technology continues to evolve.

Side-channel sensing represents a promising avenue for creating a future where smart devices are genuinely personalized and adaptable, catering to the diverse needs and interests of individuals worldwide. By exploring its full potential and addressing its challenges, we can usher in a new era of innovation and user-centric technology.

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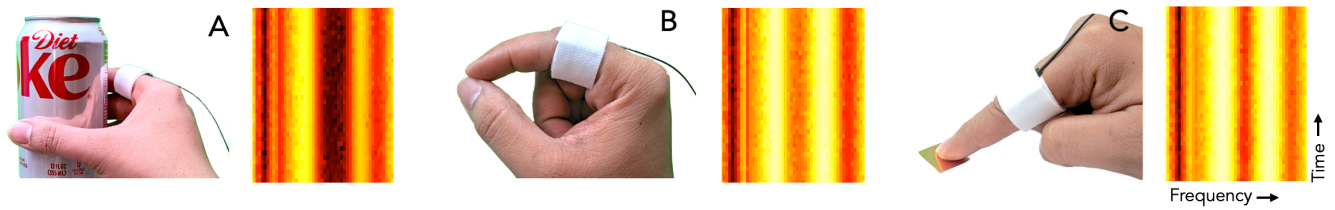


Figure 5: Z-Ring senses bio-impedance changes caused by various interactions, e.g., with objects, finger motions, and metallic surface elements like copper tape. By analyzing impedance over time and frequency, shown via spectrograms (A, B, C right), these impedance changes can drive (A) object identification, (B) gestural interactions, and (C) interaction with tangible user interface elements.

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